

SURYA REDDY NALLAMILLI

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SUMMARY

Software engineer with 3+ years in production distributed systems and LLM pipelines. Reduced P95 latency from 6s to 1.3s, cut manual underwriting review by 80% with an LLM validation pipeline, and shipped a fault-tolerant invoicing system at 500K+ invoices/month. Experienced across Java/Spring Boot, Python/Django, and AWS.

WORK EXPERIENCE

Bectran, Inc

June 2025 – Present

Software Engineer

Schaumburg, IL

- Proposed and engineered an end-to-end LLM document validation pipeline (Gemini, Java/Spring Boot, AWS) to cross-validate 10+ document types, cutting manual underwriting review time by **80%**.
- Reduced P95 latency from **6s to 1.3s** by diagnosing ORM mapping bloat, adding targeted indexes, and refactoring hotspot query paths across high-traffic workflows.
- Architected a fault-tolerant invoice delivery system (Spring Boot, ActiveMQ) processing **500K+ invoices/month** with idempotent retry logic and full audit logging.
- Designed and owned the architecture for a distributed credit bureau monitoring system across **10,000+ accounts** and 5 bureaus with zero rate-limit breaches, using a custom two-tier batch scheduler.

University of Illinois Chicago

Oct 2023 – May 2025

Full Stack Developer

Chicago, IL

- Architected a HIPAA-compliant federated research platform (Django/PostgreSQL) across 5 medical centers, enabling unified cross-site queries and improving data accessibility by **70%**.
- Delivered a 3D brain model visualization pipeline (NiiVue.js) with a cron-based scheduler and real-time UI polling, rendering volumetric models in under **3 seconds**.
- Built a DICOM anonymization pipeline with pixel-level de-identification and per-tag metadata scrubbing, reducing manual effort by **90%**.
- Optimized a React dashboard with GridStack.js configurable widgets, resolving high-volume concurrent DB call bottlenecks via server-side filtering and pagination, accelerating researcher time-to-insight by **40%**.

Cognizant Technology Solutions

August 2021 – June 2023

Software Engineer

Hyderabad, India

- Engineered a high-throughput data integration pipeline syncing **200K+ supplier records/day** between procurement and ERP systems, with schema transformation, dead-letter error routing, and retry logic.
- Reduced data defect rate by **60%** with Python/Pandas validation layers catching type mismatches, missing fields, and duplicates pre-ingestion, eliminating correction cycles that delayed supplier onboarding.
- Built a configuration-driven framework decoupling field governance from application code, cutting integration delivery time by **90%**.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, SQL, Django, Flask, Spring Boot, React, Node.js

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, SQS), Docker, Kubernetes, Nginx, Git, GitHub Actions, ActiveMQ

Databases & Data Tools: PostgreSQL, MySQL, MongoDB, Oracle, Pandas, NumPy, ELK Stack

Practices: REST APIs, Microservices, Agile/Scrum, Unit Testing, Code Review, Distributed Systems

EDUCATION

University of Illinois Chicago

August 2023 – May 2025

Master of Science in Computer Science | **GPA: 3.89/4.0**

Chicago, IL

PROJECTS

AI-Powered Incident and Root Cause Analysis Platform | *Node.js, OpenAI API*

- Developed a Node.js observability platform with time-windowed log aggregation that correlates and groups related service failures across distributed systems, reducing manual triage time by **60%**.
- Integrated OpenAI API to auto-generate structured root cause analysis reports with confidence scoring and actionable remediation steps, eliminating manual post-incident documentation.

RAG-Powered Document Intelligence System | *Python, FastAPI, OpenAI API, PostgreSQL (pgvector), Docker*

- Implemented a RAG pipeline over private document corpora using vector embeddings and pgvector, enabling semantic search and context-aware Q&A with sub-second retrieval latency.
- Exposed via a FastAPI REST endpoint with chunking, re-ranking, and hallucination guardrails, reducing incorrect responses by **40%**.